

# OpenWebSearch.eu as a Data Source for AI

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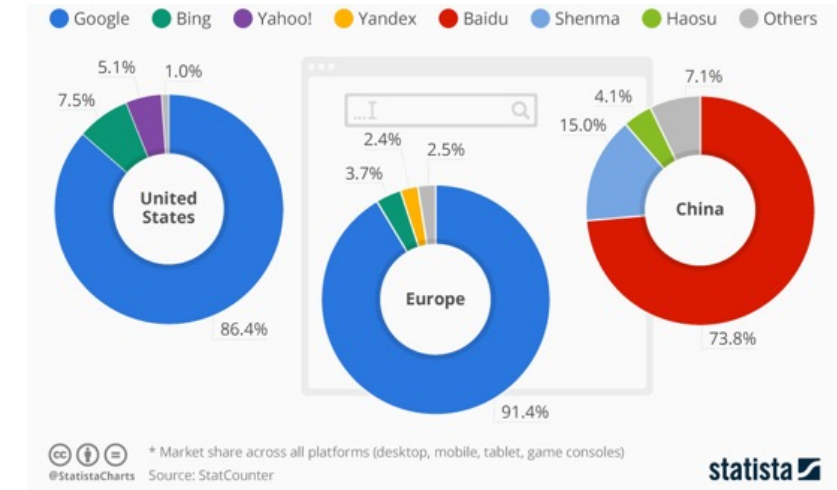


# Web Search & Web Data - a Critical Infrastructure

- A **critical infrastructure** for society — and now for AI — comparable to satellite navigation
- A **market oligopoly**: controlled by a handful of gatekeepers (Google, Microsoft, Baidu, ...)

## Consequences

- Reduced user choice
- Dependence in the information space
- Smaller researches and innovators are locked out



How to enable researchers, innovators and smaller companies to utilise the web as **data resource** for training and fine-tuning AI systems and to develop **AI-based Search Systems** without massive investments in infrastructure?

# OpenWebSearch.eu

Building an **Open Index of the Web** to lower the cost for researchers, innovators, and companies to **tap the Web as resource**

## Four objectives

1. **Open technology stack** — open pipelines for crawling, preprocessing, indexing
2. **Resource provision** — a network of providers on EuroHPC / AI Factories
3. **Added-value services** — dashboard, tools, data-access APIs
4. **Bootstrapping the ecosystem** — third-party calls, community, use-cases in **search, analytics, and AI**.

## 14 core partners

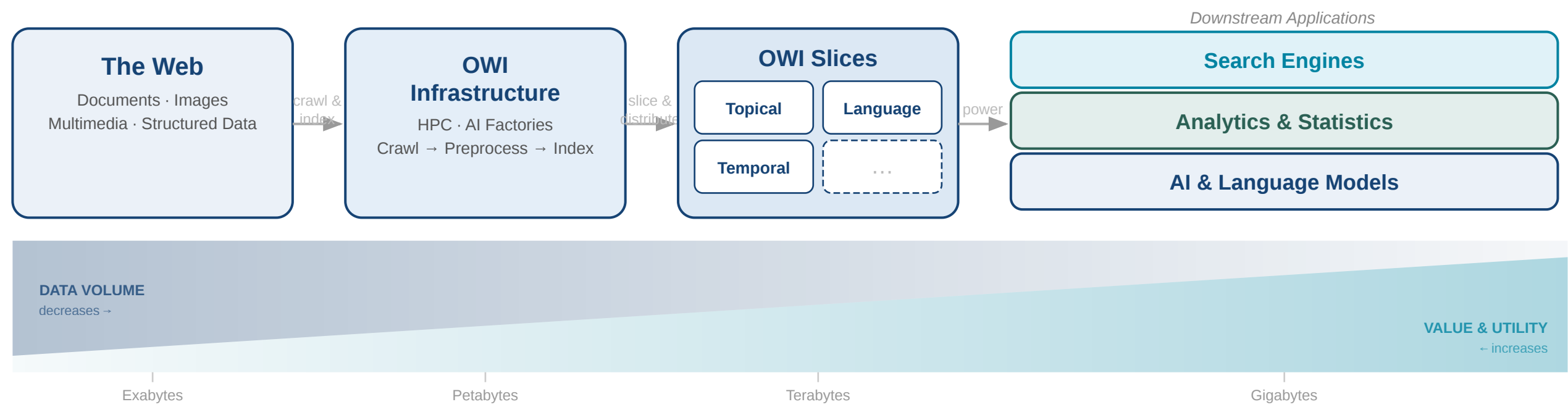


Uni Passau · LRZ · CERN · CSC · IT4I · DLR · TU Graz · Bauhaus-Uni Weimar · Radboud · Uni Leipzig · A1 Telekom · Open Search Foundation · NLnet · SUMA-EV — plus 16 funded third-party projects.

# The Goal: An Open Web Index

A **web index** is the data structure behind every search engine — fast, query-based access and ranking over web documents.

**Our proposition:** a collaboratively created, **open and transparent** web index, running on Europe's HPC centres and AI Factories.



# Building the Index

## Federated HPC Workflow

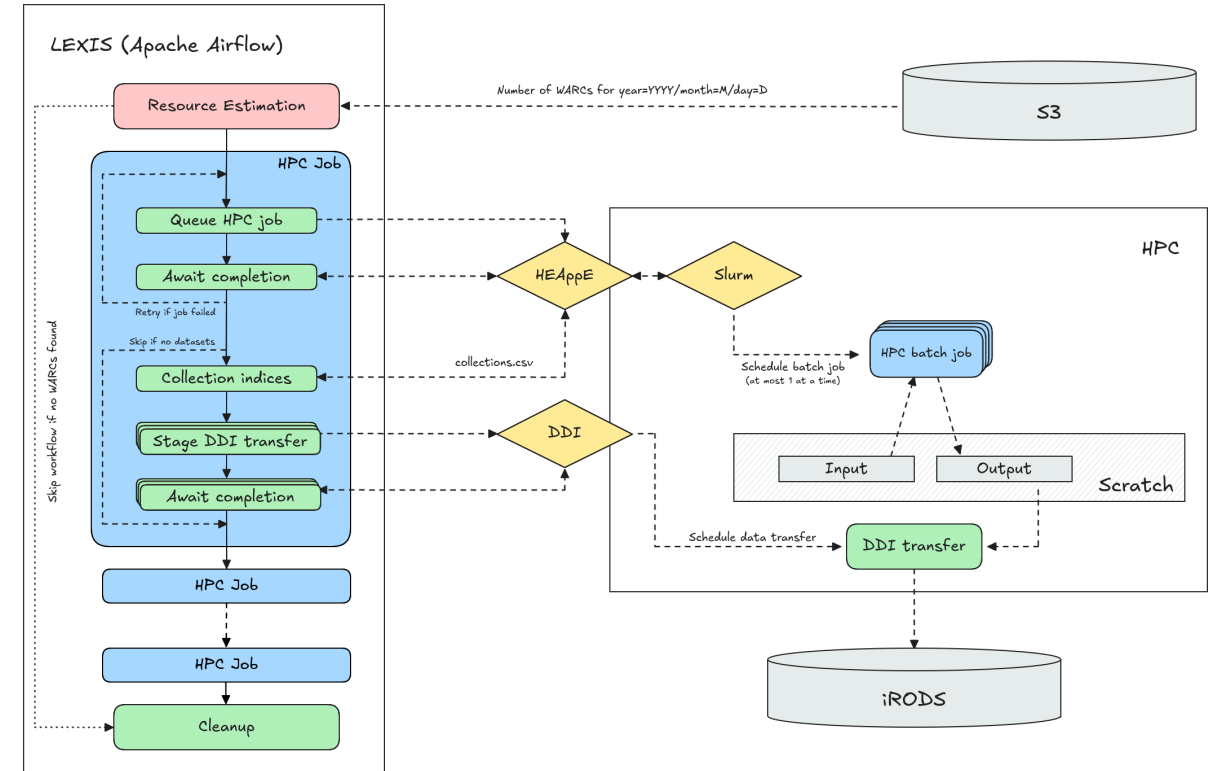
- Continuous Crawling on Cloud Partitions
- Daily Batch HPC Workflows: WARC processing · indexing · semantic enrichment, running
- Runs federated at **LRZ** (Garching) · **IT4I** (Ostrava) · **CSC** (Helsinki)

## Federated storage

- Object Storage for Crawl Data
- File Storage for Index Provision
- SQL and slow full-text search for file storage based slice and dice

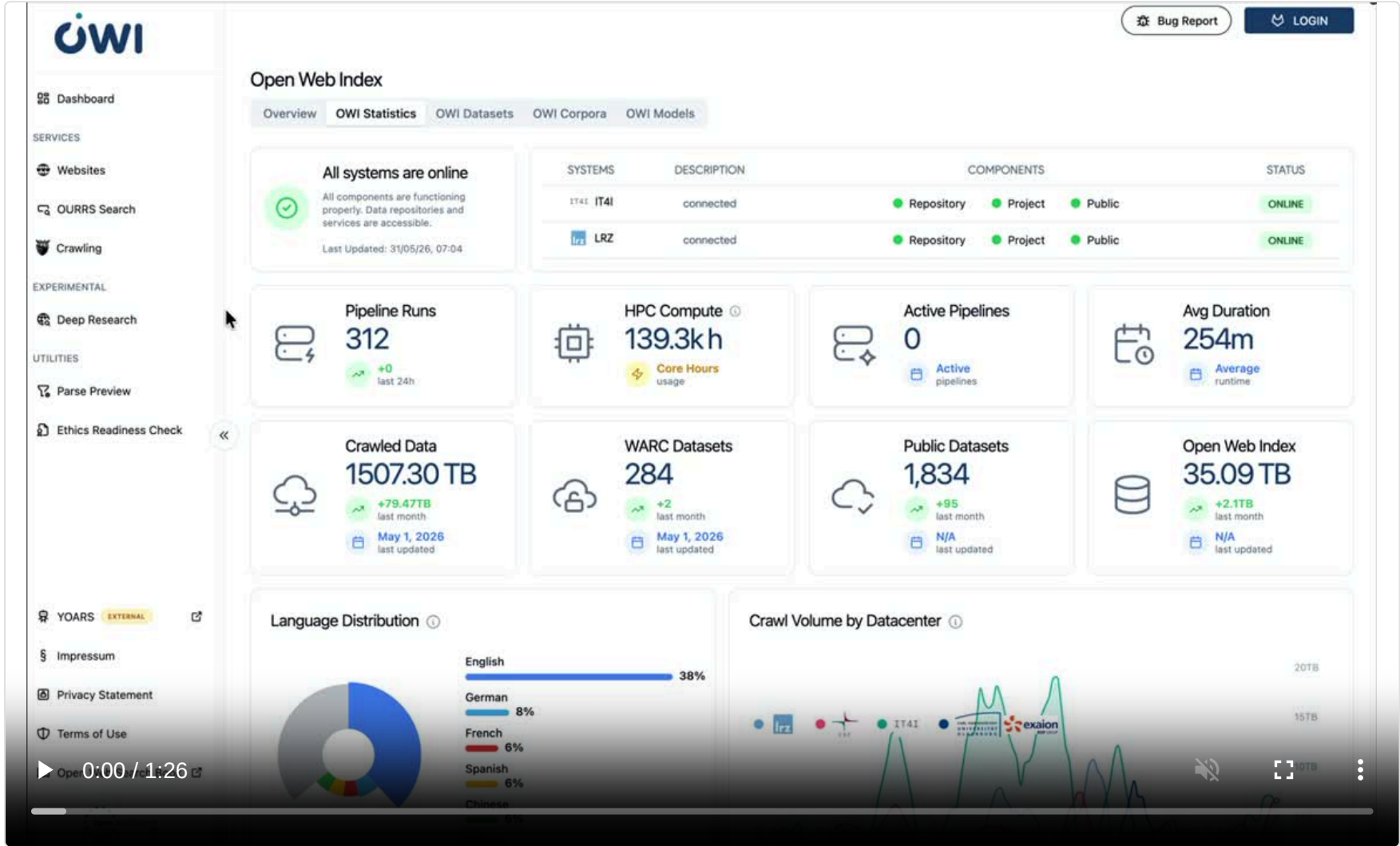
## Orchestration

- LEXIS (Apache Airflow) orchestrates jobs across schedulers
- stages data transfers, aligns daily output across all centers



LEXIS / Apache Airflow: federated OWI pipeline orchestration

# The Open Web Index Today



1,507 TB

data crawled

35 TB

Open Web Index

1,834

public datasets

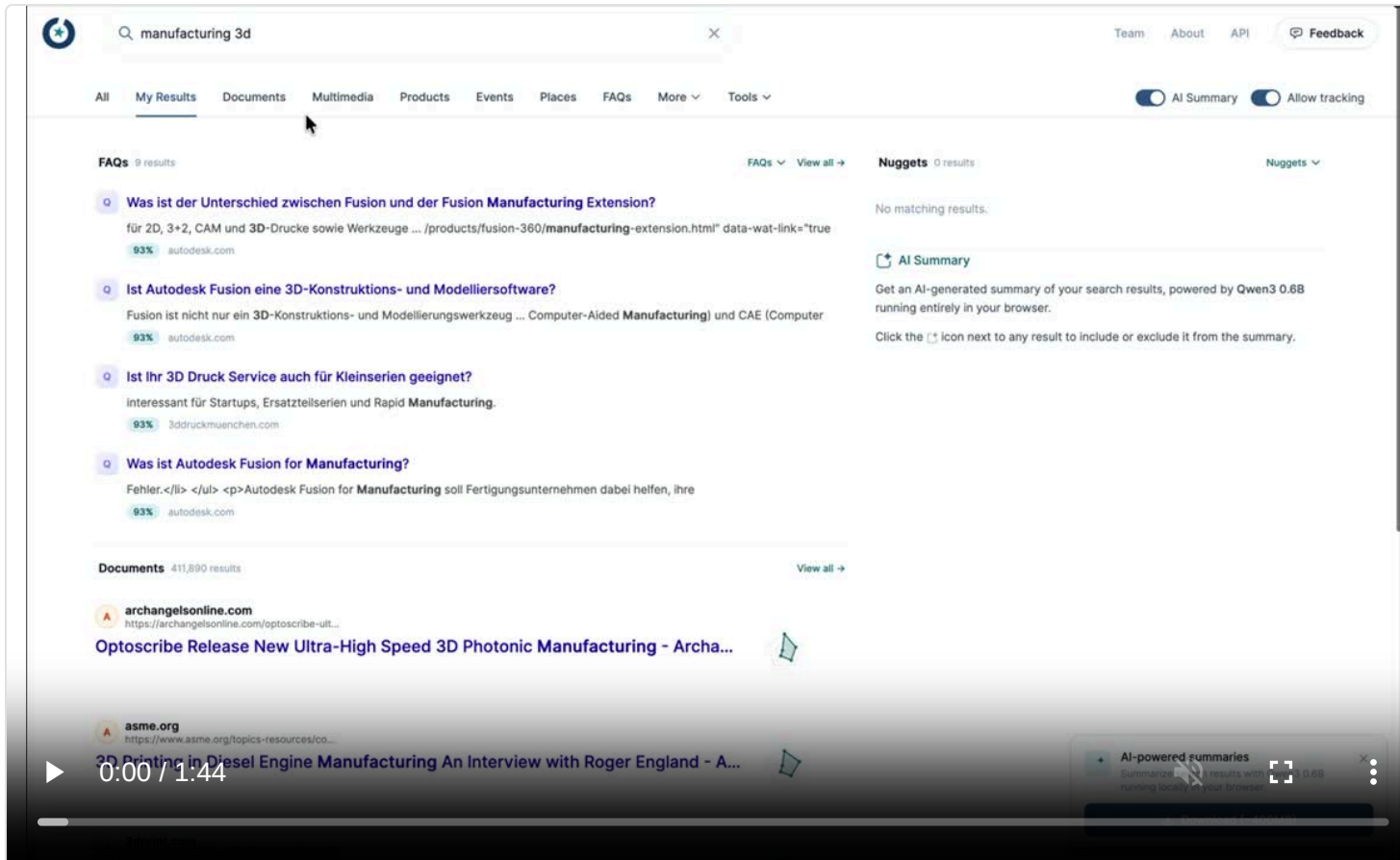
284

WARC datasets

Continuously refreshed — +79 TB crawled and +95 datasets in the last month alone.



# OUR<sup>2</sup>S Web Search Engine



**2 Months of data**

04/25 + 05/25

**598.1M Pages**

**453.4M Entities**

**14 verticals**

documents · media · products · events · places · FAQs ·  
Scholarly · Artefacts · Actions · Pages · Sites · Persons ·  
Conversations · Courses

**in-browser AI**

private Qwen3 summaries, ~400 MB

OUR<sup>2</sup>S turns the OWI into a **working search engine** — documents *and* structured entities, summarised privately in the browser.



# Web data for training & fine-tuning

# Modalities: The Index as a Multimodal Filter

## MEDIA ASSETS IN ONE MONTH OF THE OPEN WEB INDEX

**600M**

web documents

**142.1M**

media assets

**127.8M**

images

**4.9M**

videos

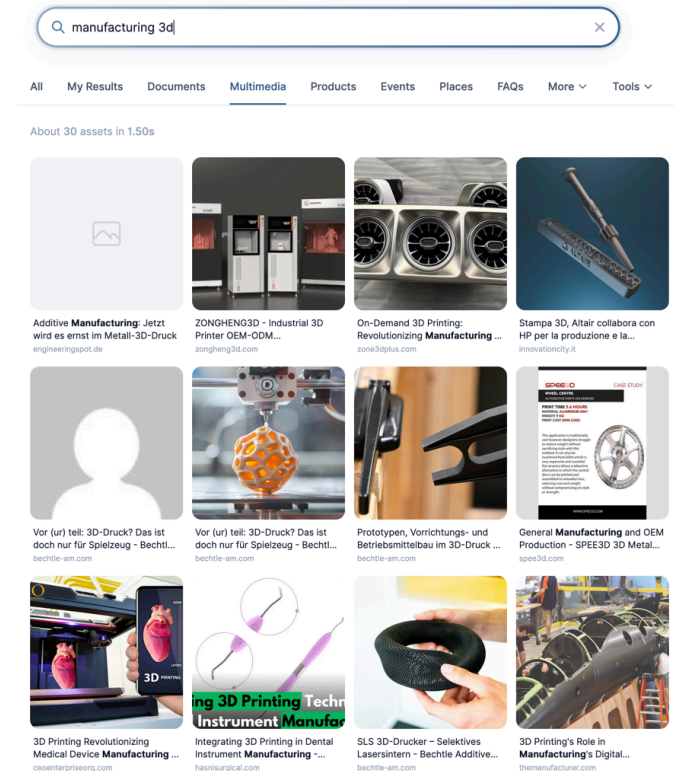
**532k**

other media

**208k**

audio

The index lets you **select training data** by modality, language, quality, and host — *before* paying to download and process it. Storage costs drop when re-downloading text & multimedia can be **targeted and selective**.



Example: multimedia results for a manufacturing query — images, 3D models, product shots.



# Domain-Specific Data

"MANUFACTURING" DOCUMENTS IN A SINGLE MONTH OF THE OPEN WEB INDEX

**6.3M**

"manufacturing"

**2.5M**

"CAD"

**996k**

"spare parts"

**583k**

"3D printing"

**498k**

"datasheet"

**Also present:** CNC machining (150k) · injection molding (120k) · industrial automation (562k) · robotics (788k) · PLC (782k) — a domain corpus you can filter out of the OWI.

Keyword matches over one month, English-leaning — a **candidate pool** before curation, dedup, and licensing.

Manufacturing is **just one example** — the OWI holds rich, filterable corpora for *any* domain (law, health, science, finance, engineering, ...), **millions of documents per month**.







# Structured Web Data

OWI parses **schema.org / Microdata** and classifies pages — including **forums and Q&A** — into **453.4M typed entities**, with three kinds of value:

## ~83M Deeper knowledge

articles (79.4M) · scholarly (897k) · books (1.7M) ·  
datasets · courses · medical  
→ **training & RAG**

## ~39M Behavioural

blog posts (18.2M) · questions (17.5M) · Q&A pages  
(717k) · reviews · social posts · forums  
→ **instruction & evaluation**

## ~120M Search & agents

products (91M) · persons (10M) · organisations (6.2M) ·  
events (5.2M) · search actions (3.7M)  
→ **agentic search & affordances**

The same crawl yields **knowledge to learn from, behaviour to learn from, and entities to act on.**

# Search becomes the memory layer

What lives in the weights, and what stays retrievable?

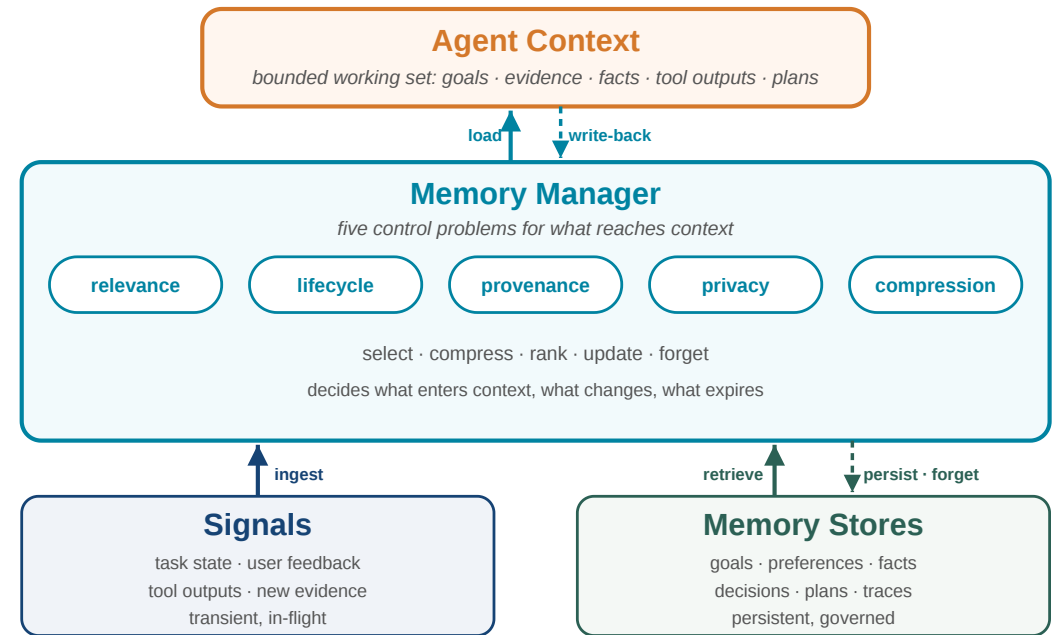
# Search as Memory: Parametric vs. Non-Parametric

## Retrieval-Augmented Generation (RAG)

Search for context relevant documents.  
Usually as Vector search, to cover linguistic and semantic diversity.

**RAG introduces a "memory" hierarchy:**

- **parametric:** model capabilities & internal knowledge
- **non-parametric:** external knowledge



RAG is essential for freshness, domain specific adaption and data privacy, but requires scalable and efficient **search infrastructures** in an AI system

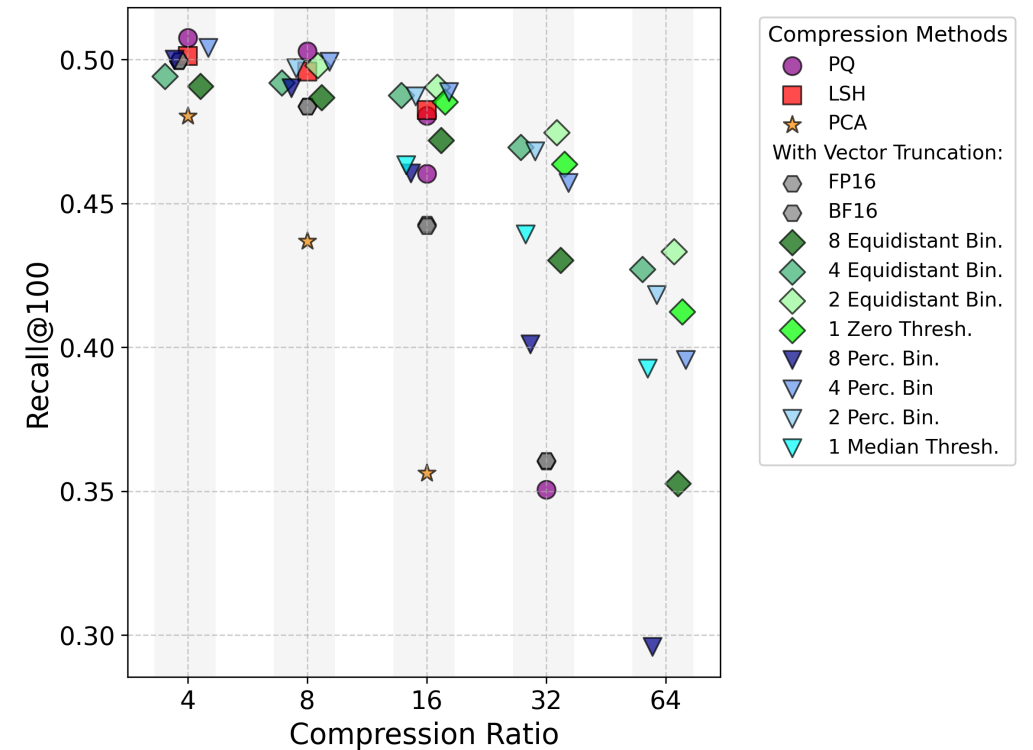
# Web-Scale Vector Memory Has a Cost

The Open Web Index provides **embeddings** for retrieval on selected web documents, but increases compute and storage significantly.

## Why compression matters

- Compute is manageable / scalable
- But embeddings are **expensive to store and serve** ( $O(n)$  for exact queries)
- compression trades index size against retrieval quality
- Quantisation wins: results hold recall while compressing **16–32×**

Compression essential for **scaling up search**.



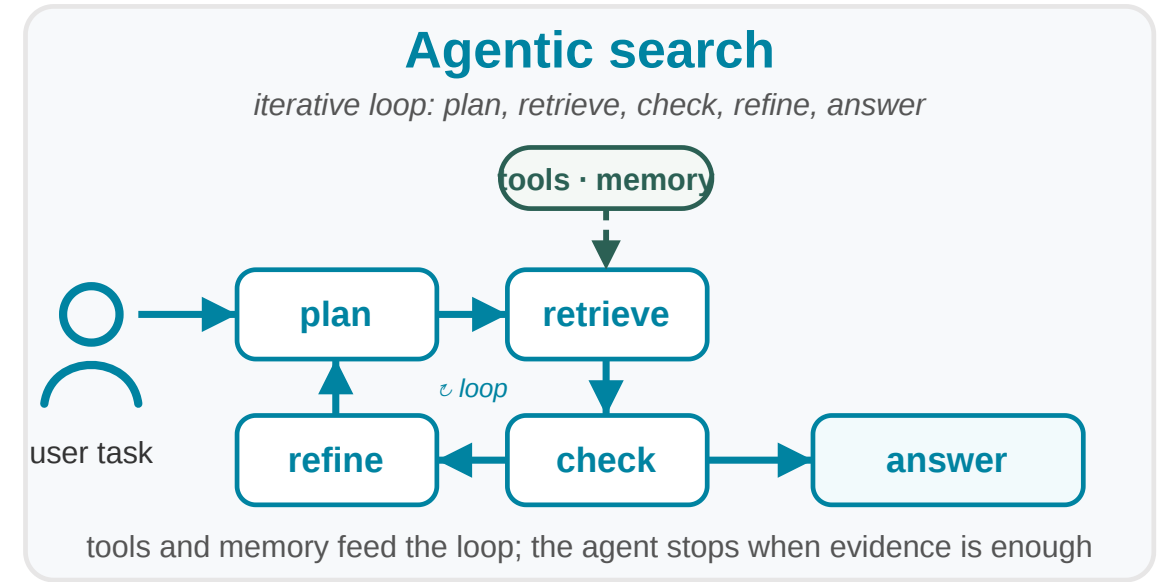
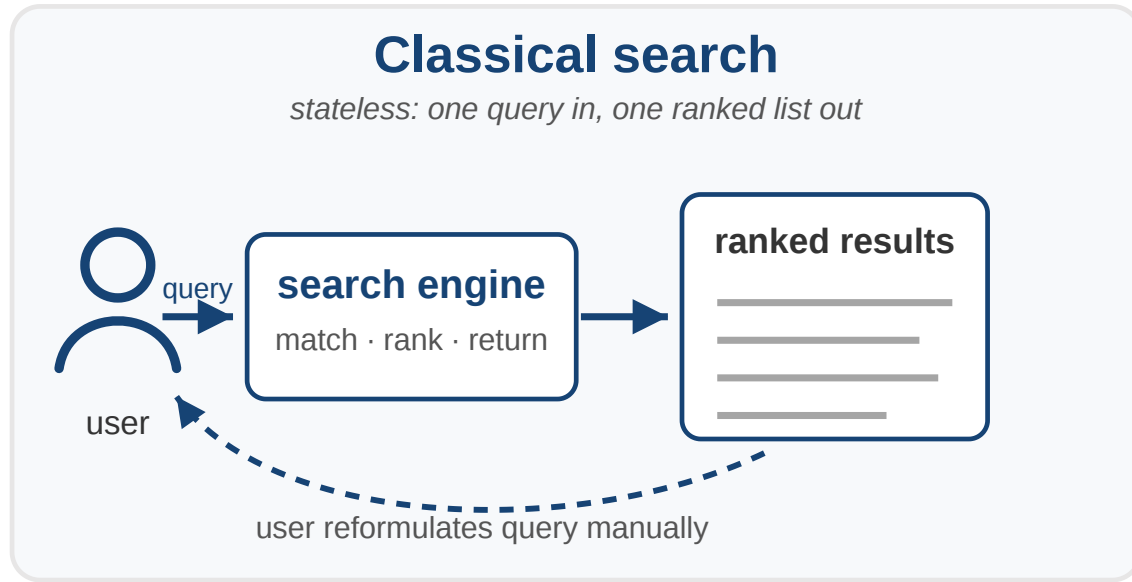




# From retrieval to agents

Retrieval moves into a control loop — and needs actions and traces.

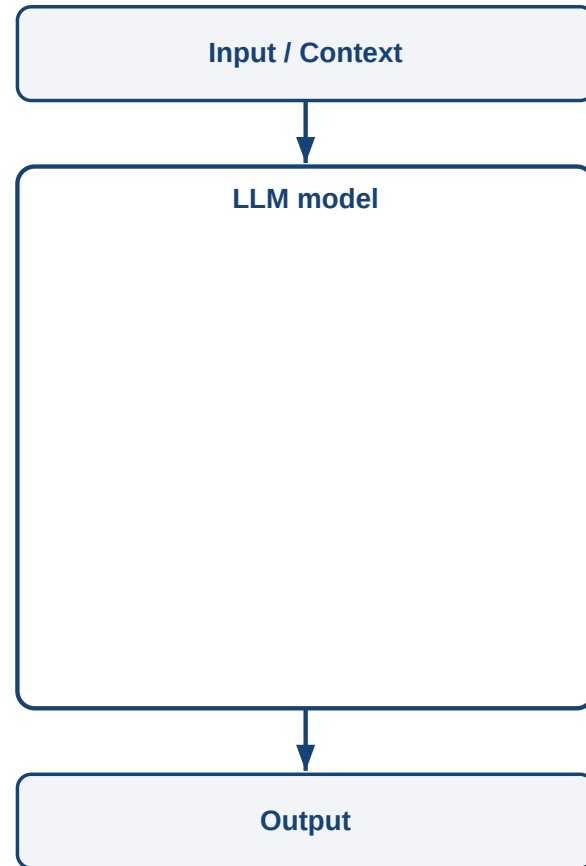
# Agentic Search: Retrieval in a Control Loop



Agents do not just **consume** search results — they **control** retrieval: plan, inspect evidence, call tools, reformulate, synthesize with provenance.

*Related paradigms: ReAct (arXiv:2210.03629); Toolformer (arXiv:2302.04761); Self-RAG — adaptive, self-reflective retrieval (arXiv:2310.11511).*

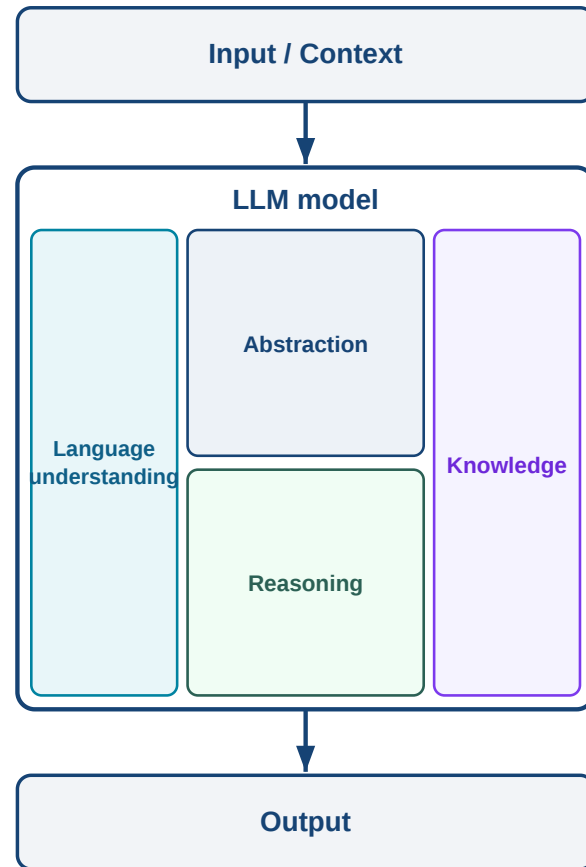
# A System View of Agentic Search



How much knowledge must sit in **weights** vs. **external memory**? Externalising it → **smaller model + fresher, controllable knowledge** — trading latency & more loops for

*Memory roles after CoALA: working · episodic · semantic · procedural (Sumers et al. 2023, arXiv:2309.02427).*

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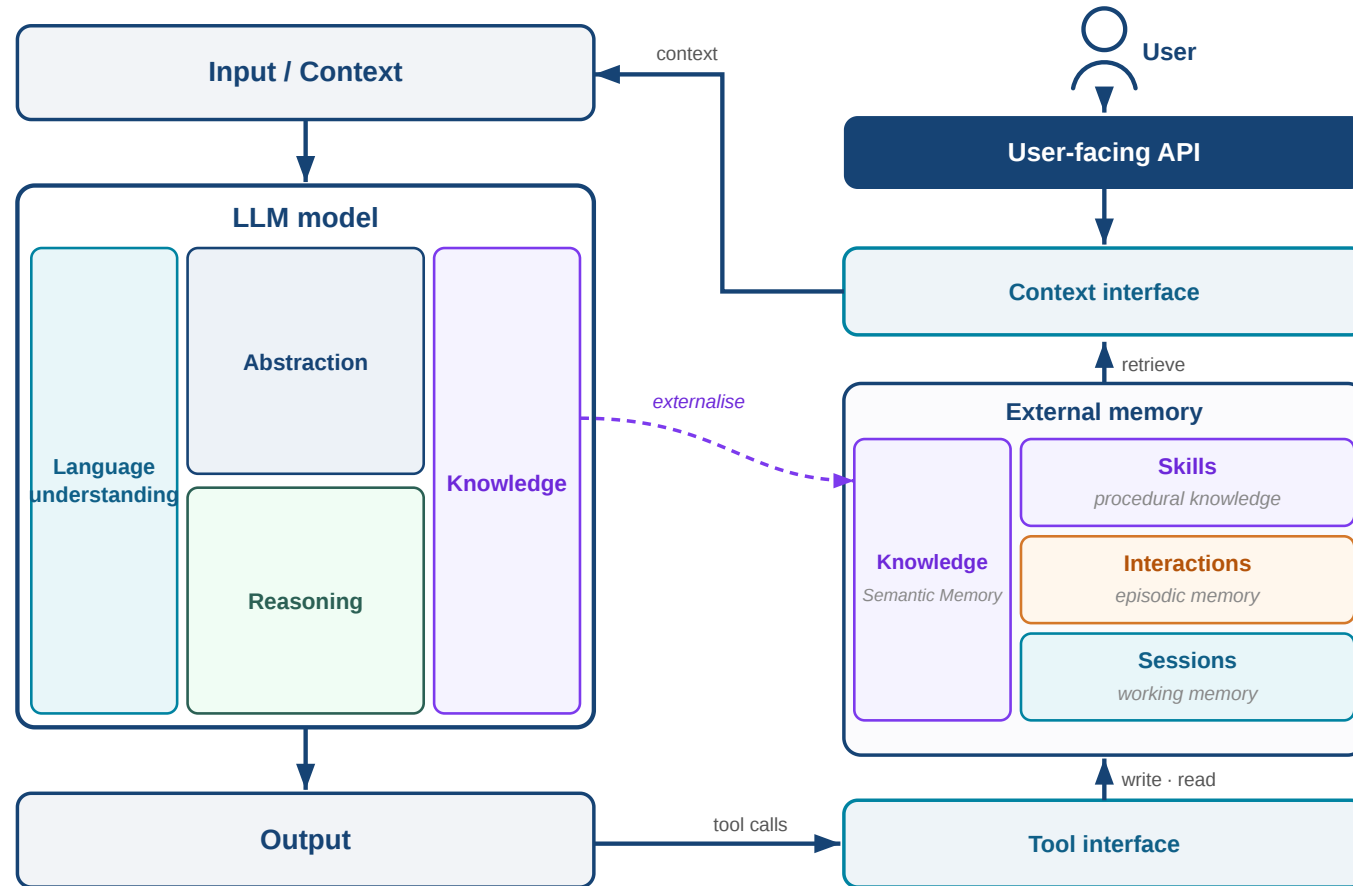


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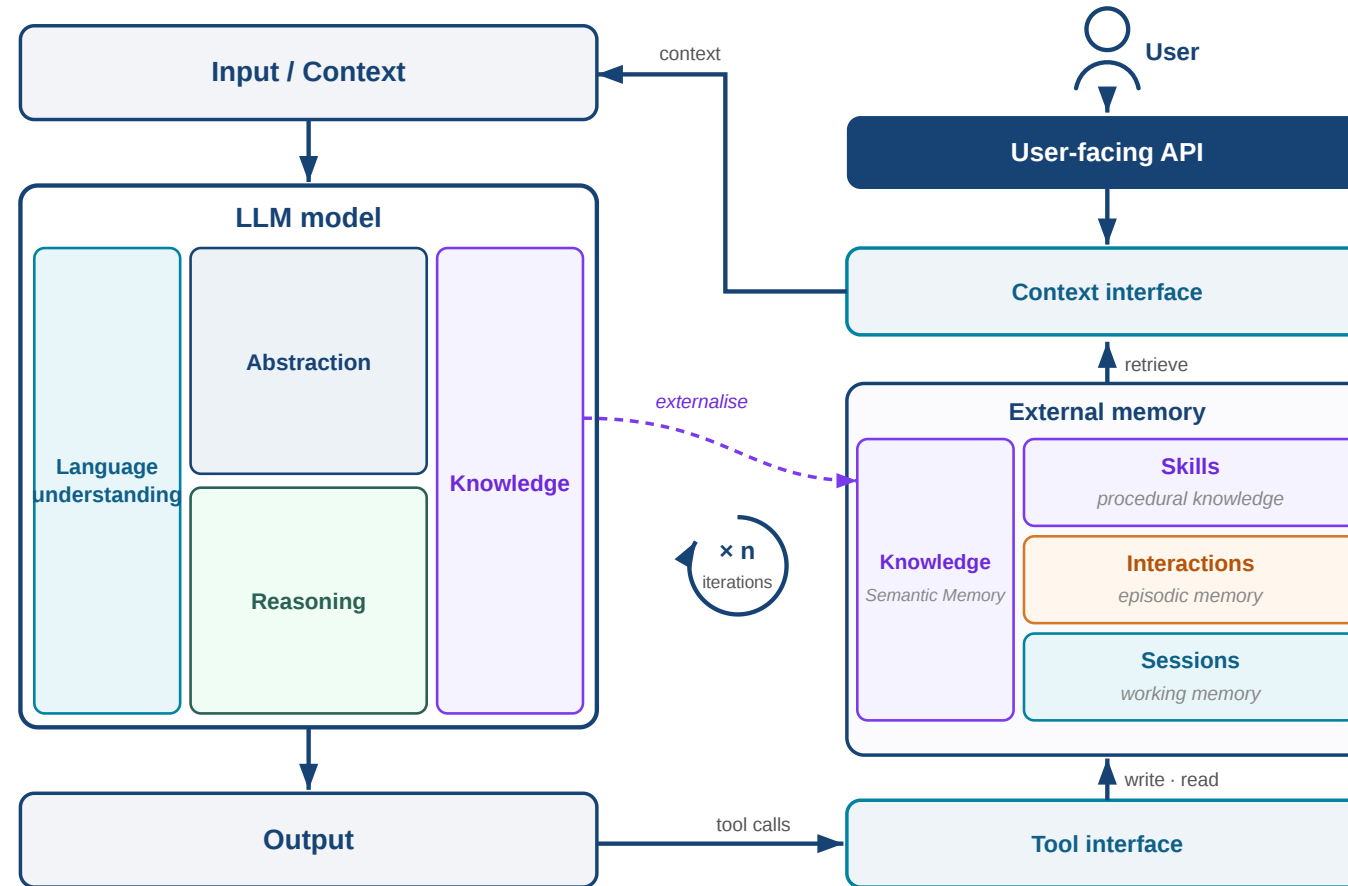
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# Structured Web Data Becomes Agent Affordance

## ACTION & APPLICATION ENTITIES IN THE OPEN WEB INDEX

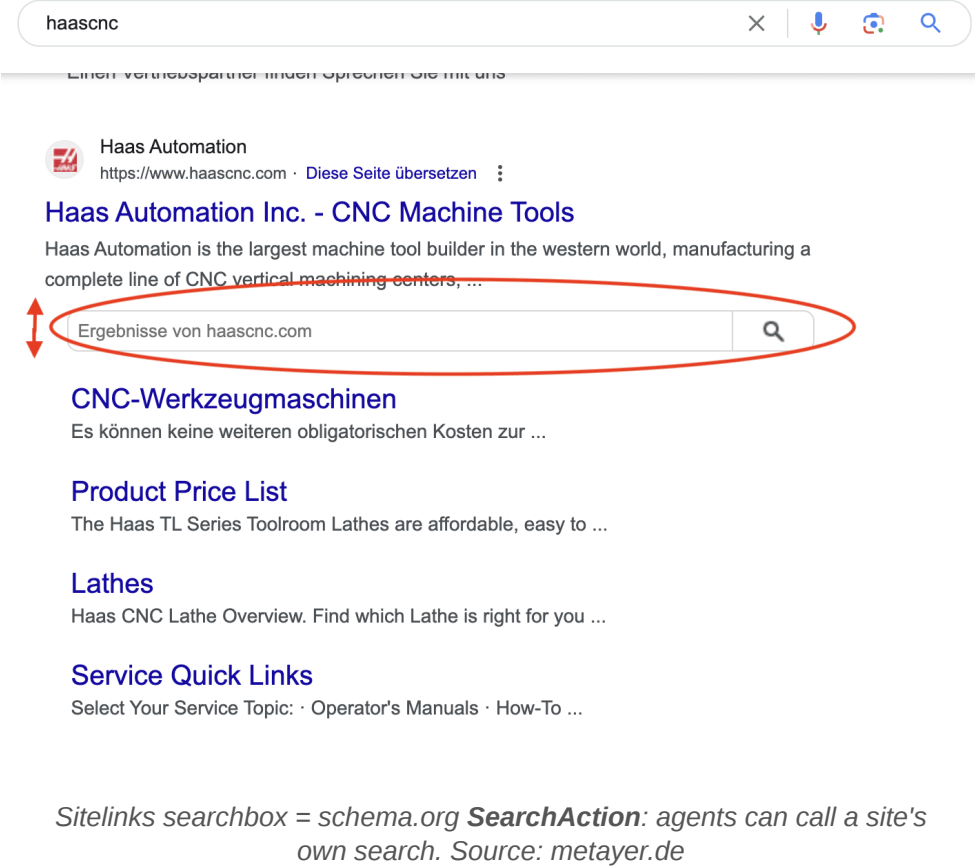
3.65M  
SearchAction

909k  
SoftwareApplication

114k  
WebApplication

94k  
MobileApplication

Agents need **discoverable actions** — what can be searched, called, installed, or configured. The index is a **candidate generator**, not blind trust.



# Beyond Web Data: Traces & Evaluation

OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

## Synthetically creating agent traces

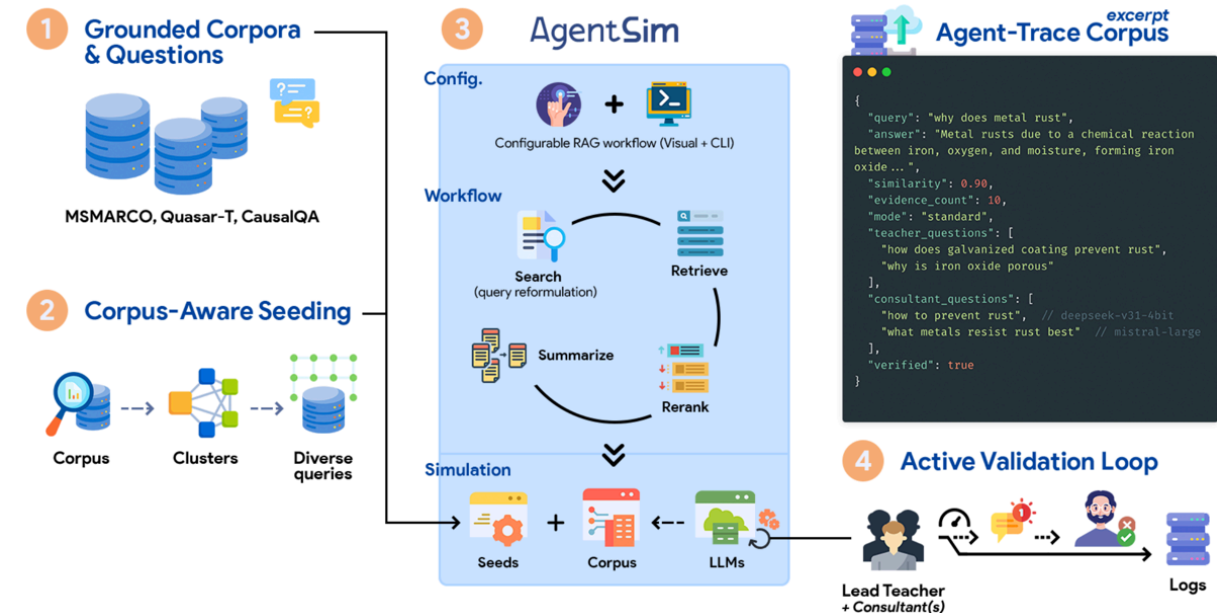
**AgentSim** — simulates RAG agents over document collections to generate synthetic, **document-grounded** reasoning traces

- **103k+** verifiable steps (Agent-Trace Corpus)
- 100% grounding rate on substantive answers.
- Agent-trace synthetic data generating is both **GPU compute heavy** and requires **search systems**.

*AgentSim, Agent-Trace Corpus · arXiv:2604.26653*

## Comparing to simulated human users

**UXSim** — grounds an adaptive LLM agent with classical simulators → accurate, **explainable** user-search simulation without costly user studies.







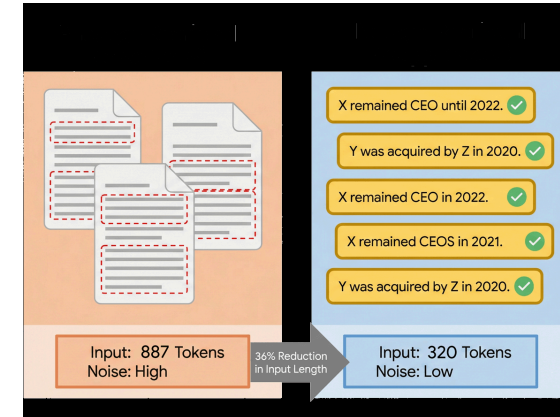
# Beyond Web Data: Knowledge & Interaction

## OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

How to structure knowledge so agents find it?

**NuggetIndex** — distil documents into atomic "nuggets": 887 → 320 tokens, far less noise, facts kept for grounding.

*Own work — nugget-level indexing for agent knowledge*



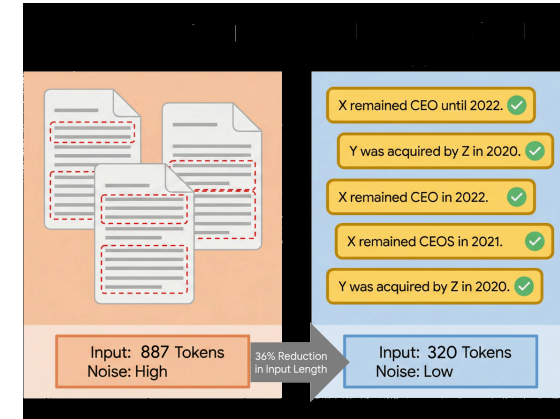
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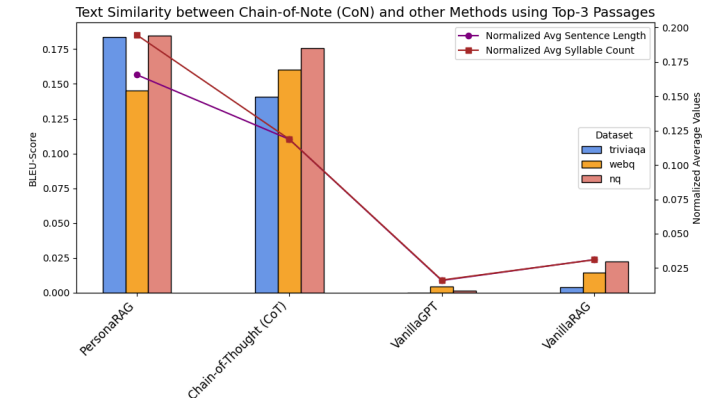
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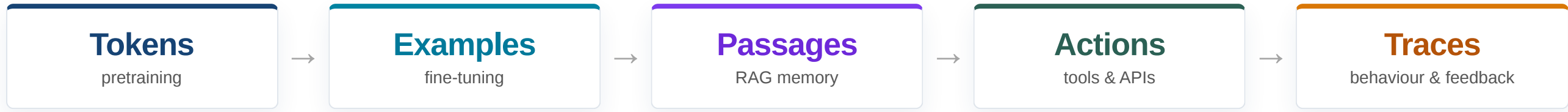
How should retrieval adapt to the user?

**PersonaRAG** — user-centric agents personalise retrieval → higher answer quality (BLEU) than CoT and vanilla RAG across TriviaQA, WebQ, NQ.

*Zerhoudi & Granitzer 2024 · arXiv:2407.09394*



# Conclusion



## What we covered

- The web is an open, reusable **data layer** — OpenWebSearch delivers **data + an index**
- **Filter** it for training & fine-tuning; **serve** it as RAG memory & agentic search
- Agentic search needs more than data: **interaction, knowledge structure, behaviour, evaluation**
- Developing Agentic Search is not only GPU-intensive, it requires complex systems.

## Let's cooperate

We are **looking forward to collaborations** on open web data, retrieval memory, and agentic search — research projects, datasets, and use-cases all welcome.

→ [openwebsearch.eu](https://openwebsearch.eu) · Chair of Data Science, University of Passau

# References

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